MILITARY SPECIFICATION

FUZE, PD, M536; LESS BALL ASSEMBLY; METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of fuze designated as *Fuze, PD, M536; less Ball Assembly.*

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this Specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirement for

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-D-14978	-	Detonator, Stab, M55
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

FSC: 1390

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STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9252295 - Fuze, PD, M536

PUBLICATIONS

ARMY

ET-9252295 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Fuze.-The fuze shall comply with all requirements specified on Drawing (dwg.) 9252295 and with all requirements specified in applicable specifications.

3.3 Spring embrittlement.-The springs shall not break when pulled to the minimum point of permanent distortion. Applies only when springs are electroplated after forming.

3.4 Qualitative spot test for tin.-The lock spring and rotor spring shall show evidence of tin when tested as specified in 4.4.19.

3.5 Jolt and Jumble.-The primary fuze assembly and auxiliary fuze assembly when tested as specified in 4.4.22, shall withstand the requirements of MIL-STD-331 without becoming hazardous to anyone handling them.

3.5.1 If a failure of skirt or ball occurs in the jolt and jumble tests, the item shall be considered a "no test" item and shall not reject that particular fuze assembly.

3.6 Transportation vibration.-The primary fuze assembly and auxiliary fuze assembly shall be safe and operable after test specified in 4.4.23.

3.7 Waterproofness.-A continuous stream of air bubbles shall not emerge from the primary fuze assembly and/or auxiliary fuze assembly when tested as specified in 4.4.24.

3.8 Functioning.-The primary fuze assembly shall function high order at initial impact and the auxiliary fuze assembly shall eject (ball must separate from skirt without flaring the skirt) and function high order when tested as specified in 4.4.25.

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3.8.1 Safety.-There shall be no premature burst (see 6.6) or evidence thereof in any of the test firings when tested as specified in 4.4.25.

3.9 Non-functioning.-The primary fuze assembly and auxiliary fuze assembly shall not function when tested as specified in 4.4.26. A fuze has functioned if the web of the primary fuze adapter is broken.

3.10 Contamination and distortion of Lexan parts.-Extreme care should be taken throughout all phases of manufacture to avoid contacting "Lexan" material. Specific information on "Lexan" destructive solvents may be obtained from the General Electric Co., Chemical and Metalurgical Division, Pittsfield, Mass.

3.11 Finishes.-Finish on all piece parts shall be 125 on all applicable drawings unless otherwise specified.

3.12 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents.

3.13 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.5)

3.14 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed

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necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2 (see 6.1). The first article sample shall consist of the following:

a. Three hundred thirty-four (334) of each primary fuze assembly and auxiliary fuze assembly.
b. One hundred ten (110) of each sub-assembly.
c. One hundred of each piece part.
d. Three (3) plated hammerweight assemblies or separate specimens for salt spray.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will

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be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.2.1 Quantities for first article inspection.-After classification of defects and other inspections have been performed as specified in 4.2.2, the following quantities shall be subjected to the indicated tests:

4.2.2.1.1 Torque of pinion assembly.-The one hundred ten (110) pinion assemblies shall be subjected to this test.

4.2.2.1.2 Torque of verge assembly.-The one hundred ten (110) verge assemblies shall be subjected to this test.

4.2.2.1.3 Non-arming of centerplate assembly.-The one hundred ten (110) centerplate assemblies shall be subjected to this test.

4.2.2.1.4 Arming of centerplate assembly.-After completion of test specified in 4.2.2.1.3, the one hundred ten (110) centerplate assemblies shall be subjected to this test.

4.2.2.1.5 Push out test of hammerweight assembly of centerplate assemblies.-Thirty-seven (37) centerplate assemblies and the hammerweight assemblies on each of the three legs of each assembly shall be subjected to this test.

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4.2.2.1.6 Separation test of hammerweight pin of centerplate assembly.-Thirty-seven (37) centerplate assemblies and the hammerweight pin on each of the three legs of each assembly shall be subjected to this test.

4.2.2.1.7 Retention check of firing pin spring on center plate assembly.-The one hundred ten (110) centerplate assemblies shall be subjected to this test.

4.2.2.1.8 Security of stake on hammerweight assembly.-The one hundred ten (110) hammerweight assemblies shall be subjected to this test.

4.2.2.1.9 Arming delay time and non-arming of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.10 Non-arming of centrifugal lock of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.11 Arming of centrifugal lock of housing assembly.-After completion of test specified in 4.2.2.1.10 the one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.12 Push test of retainer out of housing assembly.-The one hundred ten (110) housing assemblies shall be subjected to this test.

4.2.2.1.13 Push test of simulated setback pin into retainer set-back pin.-The one hundred (100) setback pin retainers shall be subjected to this test.

4.2.2.1.14 Salt spray.-The three (3) plated parts or separate specimens shall be subjected to this test.

4.2.2.1.15 Qualitative spot test for tin.-Five (5) rotor springs and ten (10) lock springs shall be subjected to this test.

4.2.2.1.16 Stake of protector booster on auxiliary fuze.-The one hundred ten auxiliary fuzes shall be subjected to this test.

4.2.2.1.17 Strength of web on housing, auxiliary fuze.-The one hundred ten (110) separation charge assemblies shall be subjected to this test.

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4.2.2.1.18 Qualitative spot test for cadmium (when applicable).-Five (5) rotor springs and ten (10) lock springs shall be subjected to this test.

4.2.2.1.19 Jolt.-Twelve (12) primary fuze assemblies and twelve (12) auxiliary fuze assemblies shall be subjected to this test.

4.2.2.1.20 Jumble.-Twelve (12) primary fuze assemblies and twelve (12) auxiliary fuze assemblies shall be subjected to this test.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Remaining first article inspection samples

4.2.4.1 Waterproofness of auxiliary fuze assembly, prior to or after assembly of separation charge assembly.-Seventy-five (75) auxiliary fuze assemblies, prior to or after assembly of separation charge assembly, shall be subjected to this test. If two (2) or more of the fuze assemblies fail to comply with the applicable requirement, the first article sample shall be rejected and the contractor shall take the necessary corrective action and submit a new first article sample or portion thereof, as specified by the Contracting Officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as the sample successfully passes the inspection specified or until otherwise directed by the procuring activity. NOTE: If separation charge assembly is already assembled, the separation charge assembly shall be cut out before performing this test.

4.2.4.2 Transportation vibration.-The three hundred (300) of each primary fuze assembly and auxiliary fuze assembly shall be subjected to this test.

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4.2.4.3 Waterproofness of primary fuze assembly.-Seventy-five (75) primary fuze assemblies, after completion of test specified in 4.2.4.2, shall be subjected to this test. If two (2) or more fuze assemblies fail to comply with the applicable requirement, the first article sample shall be rejected and the contractor shall take the necessary corrective action and submit a new first article sample or portion thereof, as specified by the Contracting Officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as the sample successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.2.4.4 Functioning.-Three hundred (300) primary fuze assemblies, after completion of test specified in 4.2.4.2 and 4.2.4.3 and three hundred (300) auxiliary fuze assemblies, after completion of test specified in 4.2.4.2, shall be assembled into rounds and subjected to this test. One hundred fifty (150) rounds shall be fired to impact at approximately 200 meters from the launcher on six inch thick loose gravel pad. The remaining one hundred fifty (150) rounds shall be fired at 150 feet from the launcher on six (6) to eight (8) inches (") of sand placed at 30 degree incline.

4.2.4.4.1 Rejection.-If the combined total of primary fuze assemblies and auxiliary fuze assemblies not function exceeds ten (10) (or more than six on one phase), or four (4) or more ejection failures of auxiliary fuze assemblies in the 150 foot test occur, or one (1) premature burst occurs, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

NOTE: In the 200 meter phase, a functioning round is considered good unless it can definitely be established that it went low order or did not function.

4.2.4.5 Non-functioning.-Ten (10) primary and auxiliary fuze assemblies shall be assembled into cartridges for which they are standard and subjected to this test. Velocity readings shall be recorded for this test. If any cartridge functions the first article inspection sample shall be rejected whether a velocity reading was obtained or not. If no velocity reading is obtained

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for any one round and the round does not function, the test shall be considered valid. If no velocity readings are obtained for more than one round and no round functions, another non-functioning test of ten (10) fuzes shall be conducted. Picatinny Arsenal, ATTN: SMUPA-ND3 shall be advised when velocity readings are not obtained for one or more rounds when it is necessary to test more than one quantity of ten (10) fuzes. The velocity shall be measured at 20 feet from the muzzle of the launcher using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE: To insure that the chronographs are read properly, the ratio of the two times shall not differ from 1.200 $\pm$ 0.005, the results shall be discarded and another round fired in its place.

Until a first article sample is accepted, the contractor is in no way authorized by the Government to resume production unless otherwise directed by the Contracting Officer.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Detonators from one interfix lot number from one manufacturer.

b. Separation charge assemblies from no more than two (2) successively produced acceptable lots from one manufacturer from one interfix lot number.

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c. Delay detonator assemblies from no more than two (2) successively produced acceptable lots from one manufacturer from one interfix lot number.

d. Primary fuzes and auxiliary fuzes represented in a given functioning test (See Para. 4.4.25) are to be considered matched sets and numbered and shipped as one combined serial lot.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Adapter or adapter (Impact) (see dwg. 8864595 or dwg. 9207924 respectively covering a detail of dwg. 8864594)

Categories	Defects	Method of Inspection	Code No. (See 6.1)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Thickness of web	Gage	01001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	01002

4.3.2.2 Pin firing (see dwg. 8864598 covering a detail of dwg. 8864596)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 Percent		
101.	Profile of firing pin point	Gage	02001
102.	Total length	Gage	02002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	02003

4.3.2.3 Pin, setback (see dwg. 8864650 covering a detail of dwg. 8864607)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Length to flange, along smallest diameter	Gage	03001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	03002

4.3.2.4 Retainer, setback pin (see dwg. 8876212 covering a detail of dwg. 8864607)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Depth	Gage	04001
102.	Diameter, retaining mounds	Gage	04002
103.	Concentricity retaining mounds diameter with center diameter	Gage	04003
104.	Center diameter	Gage	04004
Minor:	AQL 1.00 Percent		
201.	Any retaining mound missing	Visual	04005
202.	Evidence of poor workmanship (see 3.12)	Visual	04006

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4.3.2.5 Spring, firing pin (see dwg. 8864603 covering a detail of dwg. 8864596)

Categories	Defects	Method of Inspection (See Para. 4.3.3.2)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Force at .03 height, max.	Gage	05001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	05002

4.3.2.6 Protector, booster (see dwg. 8864628 covering a detail of dwg. 8864618)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Length to shoulder	Gage	06001
102.	Concentricity of small inside diameter with large inside diameter	Gage	06002
103.	Concentricity of small outside diameter with large inside diameter	Gage	06003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	06004

4.3.2.7 Housing, auxiliary fuze (see dwg. 8864619 covering a detail of dwg. 8864618)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Concentricity of small outside diameter with pitch diameter including perpendicularity	Gage	07001

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102.	Concentricity of small outside diameter with large outside diameter	Gage	07002
103.	Chamfer missing on large outside diameter	Visual	07003
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	07004

4.3.2.8 Housing assembly, prior to insertion of setback pin (see dwg. 8864607 covering a detail of dwg. 8864594)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor in armed position	Visual	08001
Major:	AQL 0.40 Percent		
101.	Excessive bulge on detonator	Visual	08002
Minor:	None defined		

4.3.2.9 Housing assembly (see dwg. 8864607 covering a detail of dwg. 8864594)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor fails to remain in unarmed position when centrifugal lock in retracted	Manual/Visual	09001
2.	Setback pin not into rotor	Probe	09002
3.	Setback pin retainer missing	Visual/Prove	09003
4.	Two (2) or more leaves do not contact setback pin (see 4.3.2.25)	Visual	09004
Major:	AQL 0.40 Percent		
101.	One (1) leaf will not contact setback pin because of damaged leaf (see 4.3.2.25)	Visual	09005
Minor:	None defined		

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4.3.2.10 Pinion assembly (see dwg. 8881055 covering a detail of dwg. 8864607)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Total length, min.	Gage	10001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	10002

4.3.2.11 Verge assembly (see dwg. 8864613 covering a detail of dwg. 8864607)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Total length, min.	Gage	11001
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.13)	Visual	11002

4.3.2.12 Auxiliary fuze assembly (see dwg. 8864618 covering a detail of dwg. 8864617)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Total length, max.	Gage	12001
102.	Large diameter of housing	Gage	12002
103.	Large outside diameter of protector booster	Gage	12003
104.	Small outside diameter of protector booster	Gage	12004
105.	Pitch diameter of thread	Gage	12005
106.	Major diameter of thread	Gage	12006
107.	Diameter of thread under cut, max.	Gage	12007

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108.	Concentricity of large outside diameter with pitch diameter of thread including perpendicularity	Gage	12008
109.	Length from forward end of threads to flange, max.	Gage	12009
110.	Length from small end to flange, max.	Gage	12010
111.	Separation can punctured, cut or dented	Visual	12011
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	12012

4.3.2.13 *Primary fuze assembly, prior to assembling adapter** (see dwg. 8864594)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Setback pin retainer missing	Visual/Probe	13001
2.	Setback pin missing or retracted (see 4.3.2.26)	Gage	13002
Major:	None defined		
Minor:	None defined		

*Fuze must be crimped immediately after inspection in same holding fixture.

4.3.2.14 Primary fuze assembly (see dwg. 8864594)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Large diameter of cover, max.	Gage	14001
102.	Inside diameter of adapter	Gage	14002
103.	Depth of adapter cavity	Gage	14003
104.	Length from cover shoulder to inside of adapter, max.	Gage	14004
105.	Concentricity of inside diameter of adapter with large diameter of cover including perpendicularity	Gage	14005

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106.	Radius missing at base of cavity on adapter	Visual	14006
107.	Finish on walls of adapter cavity improper	Visual	14007
Minor:	AQL 1.50 Percent		
201.	Total length, max.	Gage	14008
202.	Crimp missing or not fully 360°	Visual	14009
203.	Marking missing	Visual	14010
204.	Evidence of poor workmanship (see 3.12)	Visual	14011

4.3.2.15 Pallet (see dwg. 8884979 covering a detail of dwg. 8864607)

Categories	Defects	Method of Inspection (see 4.3.3.1)	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Overall length, min.	Gage	15001
Minor:	None defined		

4.3.2.16 Starwheel assembly (see dwg. 8889914 covering a detail of dwg. 8864507)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Total length, min.	Gage	16001
Minor:	None defined		

4.3.2.17 Box, fiber, inner, prior to sealing, for Fuze, Primary, PD, M536 (see dwg. 8864757)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Incorrect number of fuzes	Visual	17001

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102.	Fuze inverted	Visual	17002
103.	Fillers missing	Visual	17003

Minor: None defined

4.3.2.18 Sealed fiber box, inner, for Fuze, Primary, PD, M536 (see dwg. 8864757)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	18001
102.	Corners not blunted	Visual	18002
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	18003

4.3.2.19 Box, fiber, inner, prior to sealing, for Fuze, Auxiliary, PD, M536, (See dwg. 8864758)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Incorrect number of fuzes	Visual	19001
102.	Fuze inverted	Visual	19002
103.	Fillers missing	Visual	19003
104.	Cushions missing	Visual	19004

Minor: None defined

4.3.2.20 Sealed fiber box, inner, for Fuze, Auxiliary, PD, M536 (see dwg. 8864758)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	20001
102.	Corners not blunted	Visual	20002

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Minor:	AQL 1.00 Percent		
201.	Label missing	Visual	20003
202.	Marking misleading or unidentifiable	Visual	20004

4.3.2.21 Unsealed outer fiber box, for Fuze, Auxiliary, PD, M536 (see dwg. 8864759)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Large fold of bags facing wrong way	Visual	21001
102.	Bag torn or perforated	Visual	21002
103.	Bag improperly sealed	Visual	21003
104.	Fillers missing	Visual	21004
Minor:	AQL 0.65 Percent		
201.	Marking of bag misleading or unidentifiable	Visual	21005

4.3.2.22 Sealed outer fiber box, for Fuze, Auxiliary, PD, M536 (see dwg. 8864759)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Box damaged	Visual	22001
102.	Top improperly assembled	Visual	22002
103.	Strapping missing or improperly assembled	Visual	22003
104.	ICC Nomenclature missing or unidentifiable	Visual	22004
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	22005

4.3.2.3 Unsealed outer fiber box for Fuze, Primary, PD, M536 (see dwg. 9232348)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.65 Percent		
101.	Large fold of bags facing wrong way	Visual	23001
102.	Bag torn or perforated	Visual	23002
103.	Bag improperly seated	Visual	23003
104.	Fillers missing	Visual	23004
Minor:	AQL 0.65 Percent		
201.	Marking of bag misleading or unidentifiable	Visual	23005

4.3.2.24 Sealed outer fiber box for Fuze, Primary, PD, M536 (See dwg. 9232348)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Box damaged	Visual	24001
102.	Top improperly assembled	Visual	24002
103.	Strapping missing or improperly assembled	Visual	24003
104.	DOT (ICC) Nomenclature missing or unidentifiable	Visual	24004
Minor:	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	24005

4.3.2.25 Examination for contact of setback pin retainer leaves (see 4.3.2.9).-Each assembly in which one (1) or more leaves of the setback pin retainer assembly will fail to contact the setback pin shall be subjected to the following procedure:

1. The setback pin shall be operated.
2. Visual examination shall be conducted to determine whether leaves contact the setback pin.

If two (2) or more leaves fail to contact the setback pin, the assembly shall be classed defective. If one (1) leaf fails to contact the setback pin, the assembly shall be classed as Major defect. Assemblies subjected to examination in accordance with this procedure shall be returned to the lot.

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4.3.2.26 Examination for setback pin.-The position of the setback pin shall be classified a Critical Defect if the top of the setback pin is less than .067 inches from the exposed surface of the lower housing. (See Para. 4.3.2.13).

4.3.3 Testing

4.3.3.1 Dimensional control of cast parts, Code No. 27001.- In place of the normal sampling associated with the Classification of Defects, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production, or as a control of the cavity during production. Inspection or control of each cavity during production shall be performed on the above quantities from each cavity after production of each 500 parts. Individual cavity identification must be provided for each mold.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, that portion of production since the last control check, shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.3.3.2 Dimensional control of stamped parts, Code No. 28001.-In place of the normal sampling associated with the Classification of Defects and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check, shall be returned to the contractor for corrective action.

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All tools removed from production because of some fault, may, after reworking be returned to production, providing they pass the qualification test above. The contractor may request a change of inspection frequency, providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.3 Check test for deterioration of detonators (See Specification MIL-D-14978, Major Defect).-If the total time between original acceptance of any lot and the assembly of that lot into the fuze exceeds two (2) years, or if the detonators have been subjected to adverse conditons, however brief, at any time since previous tests, the detonator lot shall be subjected to and must pass the check test for deterioration specified in MIL-D-14978 immediately before the detonator lot is assembled into the fuze. This test shall be performed by the contractor on detonators selected by the Government Inspector at the facility assembling the detonators into the fuze. (See 6.4) Code No. 29001. The test shall be conducted at Government expense without cost to the contractor who loaded the detonator or the contractor assembling the detonator into the cartridge, and shall not consistute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the responsibility of either contractor.

The Contracting Officer shall arrange to have the contractor reimbursed for the expenses incurred in the performance of check tests.

4.3.3.4 Torque of pinion assembly (See dwg. 8881055), Major Defect, Code No. 30001.-The sampling plan for this test shall be in accordane with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

4.3.3.5 Torque of verge assembly (see dwg. 8864613), Major Defect, Code No. 31001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.6 Non-arming of centerplate assembly (see dwg. 8864596), Major Defect, Code No. 32001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

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4.3.3.7 Arming of centerplate assembly (see dwg. 8864596), Major Defect, Code No. 33001.-After completion of test specified in 4.3.3.6, the samples shall be subjected to this test with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4.

4.3.3.8 Verification of heat treat of centerplate (see dwg. 8864597), Major Defect, Code No. 34001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter H, single plan, with an AQL of 0.25 percent. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.9 Pushout test of hammerweight assembly of center plate assembly (see dwg. 8865596), Major Defect, Code No. 35001.-Forty-two (42) centerplate assemblies shall be selected from each lot and the hammerweight pin on each of the three (3) legs of each assembly shall be subjected to this test. If two (2) or more hammerweight assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4.

4.3.3.10 Separation test of hammerweight pin on centerplate assembly (See dwg. 8864596) Major Defect, Code No. 36001.-Forty-two (42) centerplate assemblies shall be selected from each lot and the hammerweight pin on each of the three (3) legs of assembly shall be subjected to this test. If two (2) or more hammerweight pins fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4.

4.3.3.11 Retention check of firing pin spring on centerplate assembly (See dwg. 8864596) Major Defect, Code No. 37001.-The sampling plan for this test shall be in accordance with Military Standard MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

4.3.3.12 Security of stake on hammerweight assembly (See dwg. 8864599), Major Defect, Code No. 38001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.9, using equipment in accordance with 4.3.4.

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4.3.3.13 Arming delay time (see dwg. 8864607), Critical Defect if assembly arms in less than 225 milliseconds, Major Defect if assembly arms between 225 and 240 milliseconds, and Non-arming of housing assembly, Code No. 39001.-Each housing assembly, less setback pin and retainer, shall be subjected to this test. Each housing assembly which fails to arm within the time limits specified on the applicable drawing shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.14 Transportation vibration test of centrifugal lock mechanism (See dwg. 8864607), Major Defect, Code No. 40001.-Twenty (20) unarmed fuze assemblies without setback pins and without detonators, shall be selected from each lot and subjected to this test. Each fuze assembly cover shall be conspiciously marked to indicate absence of setback pin and detonator. Fuze assemblies shall be placed in interplant shipping containers and subjected to this test.

At the onset of the test, the rotor shall be in the fully unarmed position, with the rotor gear in contact with a pin with a diameter of .078 - .001 inches. The centrifugal lock shall be in engagement with the star wheel. The pin shall then be removed and the housing assembly shall be subjected to the test. The amount of travel of each rotor shall be noted. If the travel of any rotor exceeds 36 degrees, the lot shall be rejected. All data concerning rotor travel shall be reported to Picatinny Arsenal, ATTN: SMUPA-ND3. The test shall be performed as specified in 4.4.11, using equipment in accordance with 4.3.4.

4.3.3.15 Non-arming of centrigular lock of housing assembly (See dwg. 8864607), Major Defect, Code No. 41001.-This test shall be conducted one hundred percent (100%). Any housing assembly which fails to comply with the requirement specified on the applicable drawing shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.16 Arming of centrifugal lock of housing assembly (See dwg. 8864607), Major Defect, Code No. 42001.-Two hundred twenty-five (225) housing assemblies shall be selected from each lot and subjected to this test. If three (3) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.13, using equipment in accordance with 4.3.4.

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4.3.3.17 Push test of retainer out of housing assembly (See dwg. 8864607), Major Defect, Code No. 43001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.14, using equipment in accordance with 4.3.4.

4.3.3.18 Push test of simulated setback pin into retainer setback pin (see dwg. 8876212), Major Defect, Code No. 44001.-One-hundred and twenty-five (125) setback pin retainers shall be selected from each heat treat batch and subjected to this test. If two (2) or more of each 125 fail to comply with the requirements specified on the applicable drawing, the batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.15 using equipment in accordance with 4.3.4.

4.3.3.19 Stake of protector booster on auxiliary fuze (See dwg. 8864618), Major Defect, Code No. 45001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.16, using equipment in accordance with 4.3.4.

4.3.3.20 Strength of web on housing, auxiliary fuze (see dwg. 8864619), Major Defect, Code No. 46001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 Percent. The test shall be performed as specified in 4.4.17, using equipment in accordance with 4.3.4.

4.3.3.21 Salt spray (see dwg. 8864599), Major Defect, Code No. 47001.-Number of separate specimens shall be selected in accordance with Specification QQ-Z-325. Rejection shall be in accordance with Specification QQ-Z-325. The test shall be performed as specified in 4.4.18, using equipment in accordance with 4.3.4.

4.3.3.22 Qualitative spot test for tin (See Paragraph 3.5), Major Defect, Code No. 48001.-Ten (10) lock springs and five (5) rotor springs shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement, the lot shall be rejected. NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one (1) roll of spring wire. The test shall be performed as specified in 4.4.19, using equipment in accordance with 4.3.4.

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4.3.3.23 Qualitative spot test for cadmium (See dwg. 8864648 and 8864653), Major Defect, (when applicable), Code No. 49001.-Five (5) rotor springs and ten (10) lock springs shall be selected from each lot and subjected to this test. If any part fails to comply with the applicable requirement, the lot shall be rejected. NOTE: For the purpose of this test, one lot of springs shall consist of parts manufactured from not more than one roll of spring wire. The test shall be performed as specified in 4.4.20 using equipment in accordance with 4.3.4.

4.3.3.24 Verification of heat treat of setback pin retainer (See dwg. 8876212), Major Defect, Code No. 50001.-After completion of test specified in 4.3.3.18, the three (3) setback pin retainers exhibiting the lowest push test force shall be subjected to this test. If any retainer fails to comply with the requirement specified on the applicable drawing, the heat treat batch represented by the sample shall be rejected. The test shall be performed as specified in 4.4.21 using equipment in accordance with 4.3.4.

4.3.3.25 Jolt (see para. 3.5), Major Defect, Code No. 51001.-Twelve (12) of each fuze assembly shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.22, using equipment in accordance with 4.3.4.

4.3.3.26 Jumble (see para. 3.5), Major Defect, Code No. 52001.-Ninety-six (96) of each fuze assembly shall be selected from each lot and subjected to this test. If any fuze assembly fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.22, using equipment in accordance with 4.3.4.

4.3.3.27 Waterproofness of auxiliary fuze assembly, prior to or after assembly of separation charge assembly, (See para. 3.7), Major Defect, Code No. 53001.-Seventy-five (75) auxiliary fuze assemblies prior to or after assembly of separation charge assembly, shall be subjected to this test. If three (3) or more of the fuze assemblies fail to comply with the applicable requirement, the lot shall be rejected. NOTE: If separation charge assembly is already assembled the separation charge be cut out before performing this test. The test shall be performed as specified in 4.4.24, using equipment in accordance with 4.3.4.

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4.3.3.28 Transportation vibration (see para. 3.6), Major Defect 54001.-One hundred fifty (150) of each fuze assembly shall be selected from each lot, placed in their regular containers and subjected to this test. The test shall be performed as specified in 4.4.23, using equipment in accordance with 4.3.4.

4.3.3.29 Waterproofness of primary fuze assembly (see para. 3.7), *Major Defect, Code No.* 55001.-Seventy-five (75) primary fuze assemblies after completion of test specified in 4.3.3.28, shall be subjected to this test. If three (3) or more fuze assemblies fail to comply with the applicable requirement, the lot shall be rejected. Prior to subsequent testing, fuze assemblies which fail this test shall be discarded and replaced with units which have been subjected to test specified in 4.3.3.28. The test shall be performed as specified in 4.4.24, using equipment in accordance with 4.3.4.

4.3.3.30 Functioning (see para. 3.8), Major Defect, Code No. 56001.-One hundred fifty (150) primary fuze assemblies, after completion of tests specified in 4.3.3.28 and 4.3.3.29 and one hundred fifty (150) auxiliary fuze assemblies after completion of test specified in 4.3.3.28, shall be assembled into rounds and subjected to this test. Seventy-five (75) rounds shall be fired to impact at apprixmately 200 meters from the launcher on six inch thick loose gravel pad. The remaining 75 rounds shall be fired to impact at 150 feet from the launcher on six (6) to eight (8) inches (") of sand placed at a 30 degree incline. If the combined total of primary fuze assemblies and auxiliary fuze assemblies not functioning exceed five (5), or two (2) or more ejection failures of auxiliary fuze assemblies in the 150 foot test occur, or one (1) premature burst occurs, the lot shall be rejected. NOTE: In the 200 meter phase, a functioning round is considered good unless it can definitely be established that it went low order or did not function. The test shall be performed as specified in 4.4.25, using equipment in accordance with 4.3.4.

4.3.3.31 Non-functioning (See para. 3.9), Code No. 57001.-Ten (10) primary and auxiliary fuze assemblies with inert booster and inert loaded ball shall be selected from each lot, assembled into cartridges for which they are standard and subjected to this test. Velocity readings shall be recorded for this test. If any cartridge functions, the lot shall be rejected, whether a velocity reading was obtained or not. If no velocity reading is obtained for any one round and the round does not function, the test shall be considered valid. If no velocity readings are obtained for more than one round, and no round functions, another non-functioning test of ten (10) fuzes shall be conducted.

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Picatinny Arsenal; ATTN: SMUPA-ND3 shall be advised when velocity readings are not obtained for one or more rounds when it is necessary to test more than one quantity of ten (10) fuzes. The velocity shall be measured at 20 feet from the muzzle of the launcher, using two sets of screens, each set of screens having an electronic counter. One set of screens shall be placed 14 and 26 feet from the launcher and the other set 15 and 25 feet from the launcher. NOTE: To insure that the chronographs are read properly, the ratio of the two times shall not differ from 1.200 by more than plus or minus (±) 0.005. If the ratio differs from 1.200 ± 0.005, the results shall be discarded and another round fired in its place. The test shall be performed as specified in 4.4.26, using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9252295 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Torque of pinion assembly.-The pinion assembly shall be placed in a fixture as shown on the Equipment List and the amount of torque specified on dwg. 8881055 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.2 Torque of verge assembly.-The verge assembly shall be placed in a fixture as shown on the Equipment List and the amount of torque specified on dwg. 8864613 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.3 Non-arming of centerplate assembly.-The assembly shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8864596 shall be applied to the firing pin. The maximum firing pin travel shall be noted after rotating the assembly at the specified non-arming speed. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Verification of heat-treat of centerplate.-The centerplate shall be mounted in a steel ring similar to a cover (dwg. 8861218). An axial push force shall be applied evenly over a circular area concentric to the axis of the firing pin hole, not to exceed .250 inch in diameter. The amount of deflection must not exceed that allowed by the drawing when the load as specified on the applicable drawing is applied. Other approved methods may be used. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.5 Arming of centerplate assembly.-The assembly shall be placed in a fixture and the force applied as specified in 4.4.3. The maximum firing pin travel shall be noted after rotating the assembly at the specified arming speed. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.6 Push out test of hammerweight assembly of centerplate assembly.-The centerplate assembly shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8864596 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.7 Separation test of hammerweight pin of centerplate assembly.-The centerplate assembly shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8864596 shall be applied to the hammerweight pin. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.8 Retention check of firing pin spring on centerplate assembly.-The centerplate assembly shall be placed in a fixture as shown on the Equipment List and the force specified on Dwg. 8864596 shall be applied. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Security of stake on hammerweight assembly.-The hammerweight assembly shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8864599 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.10 Arming delay time and non-arming of housing assembly.- Any electronic timing device and an electrically operated retractable simulated setback pin shall be operated simultaneously. A light source shall be concentrated on the detonator hole in the rotor. A photo-electric cell shall not trip an electronic circuit before the non-arming time specified. For the arming time, the photo-electric cell shall trip an electric circuit by the arming time specified. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.11 Transporation vibration test of centrifugal lock mechanism.-The primary fuze assembly shall be tested in accordance with the procedure specified in MIL-STD-331. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.12 Non-arming of centrifugal lock of housing assembly.- The housing assembly shall be mounted in a centrifuge and spun about its axis of symmetry at the rotational speed specified for a minimum of 0.400 seconds. Visually check to see that the rotor has not moved from the unarmed position. Visually check to see that the rotor has not moved from the unarmed position. When the visual check provides doubtful results, a gage shall be used which simulates the maximum diameter setback pin. An unarmed housing assembly shall be capable of accepting the simulated setback pin through the upper housing and rotor gear. Method 2: The housing assembly shall be mounted and spun as specified in Method 1. Upon completion of spin, a light source shall be concentrated on the rotor through the setback pin check hole in the top plate. A photo-electric cell shall trip an electric circuit. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.13 Arming of centrifugal lock of housing assembly.-The housing assembly shall be tested as specified in 4.4.12, except that the photo-electric cell shall trip an electric circuit and a light source shall be concentrated on center of housing assembly. Other approved methods may be used. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.14 Push test of retainer out of housing assembly.-The housing assembly shall be placed in a fixture as shown on the Equipment List and the required push specified on Dwg. 8864607 shall be applied to the retainer. This test is a non-destructive test provided the retainer is restaked and visually checked 100 percent.

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4.4.15 Push test of simulated setback pin into retainer setback pin.-The retainer setback pin shall be placed in a fixture as shown on the Equipment List and the required force specified on dwg. 8876212 shall be applied to a simulated mean setback pin. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.16 Stake of protector booster on auxiliary fuze.-This test shall be performed manually and the requirement specified on dwg. 8864618 shall apply. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.17 Strength of web on housing, auxiliary fuze.-The housing shall be placed in a fixture as shown on the Equipment List and the force specified on dwg. 8864619 shall be applied. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.18 Salt spray.-The applicable test specimens shall be tested as specified in Standard FED-STD-151. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.19 Qualitative spot test for tin.-The rotor springs and lock springs shall be tested in accordance with the following procedures:

a. Prepare 50 ml of saturated solution of cacotheline by dissolving small quantities of cacotheline powder in water. The solution can be considered saturated when the addition of small amounts of cacotheline will no longer dissolve.

b. Clean the parts in acetone and place two (2) lock springs or one (1) rotor spring in a clean test tube. Add 10 drops of concentrated hydrochloric acid and allow tin to dissolve for ten (10) minutes. Then add five (5) drops of distilled water and mix well.

c. Moisten No. 42 Whatman filter paper with saturated cacotheline solution and allow to partially dry.

d. Place a drop of test solution from the test tube on damp filter paper. The development of a distinct-violet spot where the test solution was placed indicates the presence of tin.

This test is a destructive test. Parts so tested shall not be returned to the lot.

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4.4.20 Qualitative spot test for cadmium.-The rotor springs and lock springs shall be tested in accordance with the following procedure:

a. Clean the part in acetone and place one (1) rotor spring or two (2) lock springs in a clean test tube. Add 10 ml. concentrated hydrochloric acid and allow cadmium to dissolve.
b. Decant fluid into a beaker and dilute with distilled water to total 10 ml.
c. Add 50 ml. of saturated hydrogen sulfide.
d. The development of a yellow precipitant indicates the presence of cadmium.

This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.21 Verification of heat treat of setback pin retainer.-The setback pin retainer shall be subjected to the Knoop Hardness test in a accordance with ASTM Test Method D 1474-68. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.22 Jolt and Jumble.-The primary fuze assembly and auxiliary fuze assembly with inert loaded ball assembled into skirt shall be subjected to these tests in accordance with the procedures specified in Standard MIL-STD-331. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.4.23 Transportation vibration.-The primary fuze assembly and auxiliary fuze assembly packed in their regular shipping container shall be tested in accordance with procedure specified in Standard MIL-STD-331. Assemblies subjected to transportation-vibration but not expended in subsequent testing shall not be returned to the lot.

4.4.24 Waterproofness.-The auxiliary fuze assembly without separation charge assembly and the primary fuze assembly shall be immersed in water to a depth of not less than 3 inches. The chamber shall be subjected to a vacuum pressure differential of 5 psi min. for a period of not less than 15 seconds. The water shall contain the wetting agent Niaconal NR (Sodium alkyl aryl sulfonate) or equal in a concentration of approximately 1 1/2 teasponful per gallon of water. Primary fuze assemblies so tested shall be used for the test specified in paragraph 4.3.3.34. Auxiliary fuzes so tested shall be salvaged or discarded.

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4.4.25 Functioning.-The primary fuze and auxiliary fuze assembly with assembled HE booster and inert loaded ball assembled into skirt shall be assembled with a projectile for which it is standard and then to a loaded cartridge case for which it is standard. GRAVEL: The cartridge shall be fired from a launcher for which it is standard to impact at approximately 200 meters from the launcher on six inch thick loose gravel pad. The gravel pad used shall have a 1/2 inch maximum stone size. SAND: The cartridge shall be fired from a launcher for which it is standard to impact at 150 feet from the launcher on 6" - 8" of sand and placed at 30 degree incline. Observation shall be made for proper primary fuze assembly and auxiliary fuze assembly functioning and ejection of auxiliary fuze and ball assembly.

4.4.26 Non-functioning.-The primary fuze assembly and auxiliary fuze assembly with inert booster and inert loaded ball assembled into skirt shall be assembled in a projectile for which it is standard and then to a loaded cartridge case for which it is standard. The cartridge shall be fired from a launcher for which it is standard against a verticial plywood target placed at 45 feet from point of fire. Observation shall be made for non-functioning.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C, Minimum Military Packinge.-See Para. 5.2.1.

5.2 Packing

5.2.1 Level C, Minimum Military Pack.-The primary fuze assembly and auxiliary fuze assembly shall be packed as follows:

5.2.1.1 Primary fuze assembly.-The primary fuze assembly shall be packed in accordance with dwg. 8864757.

5.2.1.2 Auxiliary fuze assembly.-The auxiliary fuze shall be packed in accordance with dwg. 8864759 and dwg. 8864758.

5.3 Marking.-Marking of the cartons shall be in accordance with dwg. 8796522 and 8799112 and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.
d. Provisions for check testing of detonator (See 6.4).

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to faciliate future data collection and analysis by the Government. These code numbers are also used to correlate the characterisitics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listd on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form *SMUPA 1010*) the numbers of drawing or Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary

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for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date

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for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonator assemblies. The test shall be conducted at Government expense without cost to the contractor who loaded the detonator assemblies or to the contractor assembling the detonator assemblies into the fuze, and shall not consistute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the assembly lot was under the jurisdiction of either contractor.

6.5 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Premature burst.-A premature burst is considered to have occurred if the round functions in the gun bore or in flight.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A-795

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